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MILITARY SPECIFICATION

BLOWERS, SOOT (SHIPBOARD USE)

1. SCOPE

1.1 Scope. - This specification covers steam boiler soot blowers suitable for cleaning boiler firesides.

1.2 Classification. - Soot blowers shall be of the following types and classes, as specified (see 6.1):

Type I - [illegible]
Type II - Rotatable.
Type III - Short retractable.
Type IV - Long retractable.
Class 1 - Maximum boiler operating pressure 600 pounds per square inch (p.s.i.); maximum steam temperature 875°F.
Class 2 - Maximum boiler operating pressure 1500 p.s.i.; maximum steam temperature 975°F.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585 Boxes, Wood, Wirebound.
PPP-B-591 Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 Boxes, Wood, Cleated-Plywood.
PPP-B-621 Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 Box, Fiberboard.
PPP-T-76 Tape, Pressure-Sensitive Adhesive Paper, Water Resistant.

MILITARY

MIL-P-116 Preservation, Methods of.
MIL-B-233 Boxes, Repair Parts.
Mil-B-857 Bolts, Nuts and Studs.
MIL-S-901 Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for.
MIL-S-933 Screws, Machine, Cap and Set; and Nuts.
MIL-D-963 Drawings, Production, Procedure for Procurement of.
MIL-S-1222 Studs, Continuous Thread [illegible] Hexagon; and Steel Bars, Round- High Temperature Service.
MIL-B-10377 Boxes: Wood- Cleated, Veneer, Paper Overlaid.
MIL-L-10547 Liners, Case Waterproof.
MIL-P-15024 Plates, Identification—Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-P-15137 Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-P-15424 Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.
MIL-C-16173 Corrosion Preventive Compound, Solvent Cutback, Cold Application.

STANDARDS

MILITARY

JAN-STD-19 Welding Symbols.
MIL-STD-22 Welded-Joint Designs.
MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.
MIL-STD-167 Mechanical Vibrations of Shipboard Equipment.

FSC 4410

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MIL-STD-271 Nondestructive Testing Requirements for Metals.
MIL-STD-278 Welding and Allied Processes for Machinery for Ships of the United States Navy.

PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS 250-692-13 - Radiographic Standards for Steel Castings.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS ASSOCIATION
B16.5—Steel Pipe Flanges and Flanged Fittings

(Application for copies should be addressed to the American Standards Association, Inc. 70 East 45th Street, New York 17, N. Y.)

AMERICAN IRON AND STEEL INSTITUTE
Steel Products Manual

Application for copies should be addressed to the American Iron and Steel Institute, 350 Fifth Avenue, New York 1, N.Y.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS
SA106 - Seamless Carbon-Steel Pipe.
SA182 - Forged or Rolled Alloy-Steel.
SA217 - Alloy Steel Castings.
SA276 - Corrosion-Resisting Steel Bars.
Boiler and Pressure Vessel Code,
Section VIII - Rules for Construction of Unfired Pressure Vessels.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 29 West 39th Street, New York 18, N.Y.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Head design characteristics. - For all classes of soot blowers, head design stresses for all temperatures above 700°F. shall be based upon allowable stressed indicated in ASME Boiler and Pressure Vessel Code, Section VIII for all comparable materials. For the head materials specified for each class (see 3.5.3.6.1 and 3.5.3.6.2), fluctuations of not more than 25°F. are permissible.

3.2 Mounting flanges. - Class 1 soot blowers shall be furnished with 600 pounds per square inch (p.s.i.) ASA standard mounting flanges conforming to Publication ASA B16.5. Class 2 soot blowers shall be furnished with 1500 p.s.i. ASA standard flanges conforming to ASA Publication B16.5. All head inlets shall be flanged. If flanged inlet provides an excess opening to valve parts, an adapter mounting flange to ASA standards shall be furnished.

3.3 Materials. - Materials used in the construction of soot blowers for Naval ships shall be of the quality best suited for the purpose intended. The material used in the construction of any design shall be of the lightest practicable consistent with the strength required for safety and reliability.

3.3.1 Materials for identification and information plates shall be as specified in Specification MIL-P-15024, except that plastics and decalcomanias will not be acceptable.

3.3.2 Threaded parts. -

3.3.2.1 Thread standards. - Standards for threaded parts shall be in accordance with Handbook H28.

3.3.2.2. Standard bolts, nuts and machine screws. - Except where otherwise specified in the contract or order or approved by the bureau or agency concerned, all bolts, nuts and machine screws shall conform to Specifications MIL-S-933,

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MIL-B-857 and MIL-S-1222. Nuts and bolts on moving parts or subject to vibration during operation shall be secured by a satisfactory locking device.

3.3.2.3 Thread projection. - Approximately one-half thread shall project clear from the outer face of the nut when nuts are screwed home. Threaded position of such bolts and ends of a stud shall be not less than 1-1/2 times the bolt or stud diameter.

3.4 Component parts. - Soot blowers shall consist of the following parts:

(a) Operating mechanism (see 3.5).
(b) Element or lance (see 3.6).
(c) For type I and II soot blowers only, bearings or hangers (see 3.7).
(d) Boiler casing fitting (stuffing box, boiler casing seal (see 3.8).

3.5 Operating mechanism. -

3.5.1 Description. - Soot blower operating mechanism for each type shall consist of the following:

(a) Type I (stationary) operating mechanism shall consist of a steam admission device (valve) only.
(b) Type II (rotatable) operating mechanism shall consist of a steam head or housing including mechanical means of element rotation geared into the steam admission device.
(c) Type III (short retractable) operating mechanism shall consist of a steam head or housing including mechanical means of lance translation and rotation geared into the steam admission device.
(d) Type IV (long retractable) operating mechanism shall consist of the following:
    (1) Steam head to regulate steam admission device.
    (2) Traveling carriage to effect lance translation and rotation.

3.5.2 Type I, soot blowers (operating mechanism). - Unless otherwise specified (see 6.1), type I soot blowers shall be complete with steam valves. Steam valves shall conform to ASME and ASA standards suitable for the source of steam. A spring-loaded air relief valve of corrosion-resistant material (3/8 inch i.p.s. minimum) shall be provided to admit purging air to the element and to prevent combustion gases from entering the piping. A 3/8 inch connection shall be provided for this purpose. The air relief valve shall be connected to the pressurized air space by 1/2 inch i.p.s. tubing and joints as approved by the bureau or agency concerned.

3.5.3 Steam head (types II, III, and IV). -

3.5.3.1 Operation. - The steam head shall be designed to transmit steam to the element or lance from a fixed steam connection without steam leakage when the element or lance is being operated. It shall be designed to permit installation on the boiler so that the steam supply line to the head may be led at any angle in a plane normal to the axis of the soot blower element.

3.5.3.2 Steam control valve. - The steam control valve shall be either power operated by an air motor of the head or manually operated as specified (see 6.1). Manual operation may be either by means of a single endless chain or by a crank as approved by the bureau or agency concerned. Extension rods and gearing may be substituted, if space limitations so indicate.

3.5.3.3 Steam pressure control in head. - The method of steam admission and steam pressure control shall be as approved by the bureau or agency concerned. Means of pressure control shall be included in each head design to reduce the pressure at the element to the blowing pressure for optimum cleaning of the tube surfaces. A 3/8 inch NPT tap is to be provided in each soot blower head for insertion of a pressure measuring device. A plus or minus tolerance of 25 p.s.i. on this pressure will be permitted in the actual installation. This pressure control shall be an adjustable part of the head in order to effect the above designated steam entrance pressure. The control shall be capable of external adjustment.

3.5.3.4 Power drive. - When air-motor driven soot blowers are specified, these air motors shall operate both the steam control valves and the designated movement of the elements. The use of cast iron in the motors is not permitted. The drive shall also include the necessary pushbutton stations for actuation of the air-motors. Panel boards for mounting the pushbutton stations shall be provided. The panel boards shall have an interlocking device to prevent operation of more than one soot blower at a time, and an indicating light to indicate when soot blowers are in operation. Interconnecting piping between air motors and control panels shall not be furnished. Provision shall be made for hand operation of the soot blowers should control air be unavailable.

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3.5.3.5 Scavenging air valve. - A spring-loaded scavenging air valve (see 3.5.2) shall be provided on the soot blower head to provide purging air to the element and to prevent combustion gases from entering the soot blower head. A 3/8 inch connection on the soot blower head shall be provided for this purpose. The scavenging air valves shall be connected to the pressurized air space by 1/2 inch i.p.s. tubing and joints as approved by the bureau or agency concerned.

3.5.3.6 Head materials. - Minimum design requirements for head materials shall be as specified in 3.5.3.6.1 and 3.5.3.6.2.

3.5.3.6.1 Class 1, 600 psi operating pressure, 875°F. - Minimum Class 1 design requirements for head materials shall be as follows. Similar American Iron and Steel Institute designations conforming to the Steel Products Manual are also approved.

Head----------Carbon molybdenum alloy steel. Castings in accordance with ASME Publication SA-217.

Valve discs------Cast carbon molybdenum alloy steel in accordance with ASME Publication SA-217 or forged carbon molybdenum alloy steel in accordance with ASME Publications SA-182 or SA-276; seating area faced with ring of not less than 3/32 inch layer of cobalt alloy Stellite No. 6 or equivalent type.

Valve seats------Materials shall be the same as for the discs. Seats may be made an integral part of the body or may consist of tubular insert welded into the body. Seating area shall be faced with finished thickness of not less than 3/32 inch layer of cobalt chromium alloy Stellite No. 6 or equivalent type.

Valve stems-----Corrosion-resisting steel in accordance with ASME Publications SA-182 or SA-276.

3.5.3.6.2 Class 2, 1500 p.s.i. operating pressure 975°F. - Minimum class 2 design requirements for head materials shall be as follows:

Head----------1-1/4 percent chrome; 1/2 percent molybdenum alloy steel casting in accordance with ASME Publication SA-217, grade WC6.

Valve discs------Cast chromium alloy steel in accordance with grade WC6 of ASME Publication SA-217 or forged chromium alloy steel in accordance with ASME Publication SA-182, grade F11; seating area faced with ring of not less than 3/32 inch layer of cobalt alloy of Stellite No. 6 or equivalent type.

Valve seats------Valve seats shall be of the same material as the valve discs; to be made as above for class 1 heads; seating area faced with finished thickness of not less than 3/32 inch of cobalt alloy of Stellite No. 6 or equivalent type.

Valve stems-----Corrosion-resisting steel in accordance with ASME Publication SA-182, grade F11.

3.5.3.7 Connecting piece (swivel or extension tube). - A tubular connecting piece shall be provided between the head and the element. Corrosion-resisting material shall be used where the piece passes through the stuffing box in the head. The connecting piece shall be for the purpose of securing the element to the head during field installation in a suitable manner to insure maintenance of the proper angular position of the nozzles with respect to the operating mechanism in the head. An indicator to show the angular position of the nozzles to the operator at all times shall be attached to the gear wheel or secured by other means approved by the bureau or agency concerned. The swivel tube or the element shall be at least 6 inches longer than designed length to allow for field modification to suit installation conditions.

3.5.3.8 Head interchangeability. - All heads of like design within the same class shall have fully interchangeable internal parts. All replacement heads of types II and III shall be furnished with one 360 degree cam and trigger assembly and one cam assembly that can be modified to fit all other rotation requirements. Replacement swivel tubes or elements supplied shall provide sufficient length to allow field modification to actual boiler depth conditions (at least 6 inches). Further, for all replacement heads means shall be provided to allow attachment of either 1-1/2 inch or 2-inch elements.

3.5.4 Steam head for type II, soot blowers. -

3.5.4.1 Operation. - In addition to operation requirements of 3.5.3, type II soot blower steam

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heads shall be provided with means for rotating the element in conjunction with operation of the steam control valve. Direction of operation in a clockwise or counter-clockwise direction shall be based on boiler requirement to adequately keep header and casing-pocket areas clean, and to direct loosened soot into uptake gas paths, provided total or combined angle of rotation of the element to fully open and to fully close the steam control valve shall not exceed 60 degrees of arc.

3.5.4.2 Reduction gearing.- Reduction gearing shall be provided in the head for rotation of the elements. The gear ratio shall not be less than 7 to 1. For manually operated units designed for rotation in one direction only, an arrow shall be cast on the gear wheel, the sheave wheel or operating handwheel, whichever is most convenient for observation by the operating personnel, to indicate the correct direction of rotation for operation of the blower. For power operated units, the correct direction of rotation for the element shall be indicated on the unit wherever it is most convenient.

3.5.5 Steam head for type III, soot blowers.- All head internal repair parts shall be interchangeable with those for type II. Mechanical means for translation and rotation of the nozzle or lance in conjunction with operation of the steam valve shall be provided. Translation of the lance into the furnance shall not be less than 10 inches.

3.5.6 Operating mechanism for type IV, soot blowers.- Operating mechanism shall consist of:

(a) Steam head containing poppet valve with externally adjustable pressure control of steam admission source.

(b) Traveling carriage.- The traveling carriage mechanism shall be air-motor driven. Mechanical means shall be used to translate and rotate the lance. Manual emergency means shall be provided to allow the removal of the lance from boiler cavities in case of air-motor failure. The traveling carriage must be of rugged construction, with suitable means of supporting the lance, during operation of rotating and translating, to prevent whipping. Minimum rotation speed of the lance shall be five revolutions per minute. Minimum translation speed shall be five feet per minute. Means shall be provided to close off the wall box opening should repair of the mechanism be required during boiler operation. Suitable means shall be provided to mechanically restrain the feed tube from rotation in the steam chest. In addition, the feed tube shall be suitably ground and polished to reduce wear on the packing.

3.6 Element or lance.-

3.6.1 Operation.- Elements may be rotatable, stationary or retractable. Rotatable and short retractable elements or lances are operated by a steam head (see 3.5). Stationary elements are operated by an externally adjustable steam throttling valve of suitable capacity. Long retractable elements require a traveling carriage and enclosed poppet valve.

3.6.2 Grades.- Elements shall be of the following grades:

Grade	Maximum gas temperature, °F.
I	900
II	1500
III	1800
IV	over 1800

3.6.3 Material(elements).- Grade I and II elements shall be manufactured of seamless rolled or drawn carbon steel tubing conforming to ASME Publication SA-106. The material of the grade II elements shall be completely protected against oxidation of an aluminum coating properly impregnated, or by a chromium calorizing or plating process, the coating to be applied after installation of the nozzles. The use of grade I and grade II element materials at a lower temperature than that for which it is approved by the bureau or agency concerned will be satisfactory providing the physical properties of the material at the lower temperature are suitable for the purpose. Grade III and IV elements are to be fabricated of corrosion-resisting steel alloys, grade of alloy and need of protective coating as approved by the bureau or agency concerned. All welding of elements shall conform to Standard JAN-STD-19, MIL-STD-22 and MIL-STD-278.

3.6.4 Design.- The wall thickness of the element, when manufactured of low carbon steel, shall be at least equal to that for schedule 80. When chrome alloy steel elements are used, the thickness of the wall of the element shall be as approved by the bureau or agency concerned. The end of the element opposite the soot blower head shall be closed in a manner approved by the bureau or agency concerned. The use of a cap or plug attached by screw thread for this purpose will not be satisfactory.

3.6.5 Elements for types I and II soot blower.-

3.6.5.1 Multi-nozzle element.- Multi-nozzle element is a straight pipe with at least one row of small nozzles fitted along one side. Element may be either (a) rotatable (steam valve and gearing parts of soot blower head) or (b) stationary (steam valve only).

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3.6.5.2 Adjustment of the element.- Elements shall be fabricated to exact designed lengths for the intended boiler locations. Accurate longitudinal adjustment of the element in the field to insure correct position of the nozzles relative to the boiler tubes, when blowing, is effected by field modification (cutting and threading element or swivel tube or adjusting support bar and carriage). Couplings or flanges used between the element and the swivel or extension tube shall be threaded to a minimum such that when the element is also threaded to a minimum and in correct position at least 3/4 inch of the element will be engaged.

3.6.5.3 Multinozzle design.- Nozzles shall be designed to provide maximum steam velocity. Material of nozzles shall be equal to or better than the element material. The diameter of the nozzle throat shall be properly sized to provide the best cleaning possible considering the location of the element, tube erosion, type of tube bank, and blowing pressure available. Unless otherwise approved by the bureau or agency concerned, the ratio of the combined throat areas of all nozzles installed in a soot blower element to the internal cross sectional area of the element shall not exceed 1.2 to 1. Unless otherwise approved by the bureau or agency concerned, the maximum overall length of the nozzle shall be 5/8 inch. The nozzles shall be secured by welding, or by screw threads and welding. Unless otherwise approved by the bureau or agency concerned, all nozzles shall be so installed that the outlet end of the nozzle is flush with the outside of the element. There shall be only one row of nozzles in each element unless otherwise approved by the bureau or agency concerned.

3.6.5.4 Position of nozzles.- The spacing of the nozzles shall be as approved for each boiler class application, but in no case shall the spacing of the nozzles be greater than every third tube space of the row of tubes adjacent to the element when the axis of the element is at a right angle to the axis of the tubes; or greater than 3 inches when the axis of the element is parallel to the axis of the tubes. In cases where tubes of 2 inch diameter and larger are employed and the axis of the soot blower element is at a right angle to the axis of the tubes, the spacing of the nozzles shall not be greater than every tube space to secure a nozzle spacing of 3 to 3-1/8 inches or as near thereto as practicable. Nozzles shall be arranged to blow into the lanes between the tubes and not against the tubes, insofar as is practicable. Except as specified herein, nozzles shall be installed with their axis normal to the axis of the element. When the axis of the element is at a right angle to that of the tubes, and the element is installed between two rows of tubes, staggered with respect to each other, if the tubes opposite the nozzles are 1-1/2 inches or less from the outside of the element, the nozzles shall be angled not more than 10 degrees to prevent direct impingement of the steam on the tubes. When the axis of the element is parallel to that of the tubes and the nearest tubes are 1-1/2 inches or less from the outside of the element, the nozzles shall be angled as described above.

3.6.6 Single nozzled lance for type III soot blower.- The single nozzled lance is to be designed for projection into the furnace for use and then withdrawn into the furnace wall to protect the nozzle from reaching excessive temperatures when not in use. The nozzle in the end of the lance shall be suitably angled from normal such as to direct the steam most effectively against the boiler tube surfaces.

3.6.7 Double nozzle lance for type IV soot blower.- The lance shall be suitably ground and polished to prevent excessive binding or wear on the lance or related parts. Two opposed nozzles at an angle designed to suit the requirements shall be installed near the front end of the lance in accordance with 3.6.5.3.

3.6.8 Gas temperatures above 1800°F..- When boiler furnace locations reaching maximum gas temperatures above 1800°F. require soot cleaning of the boiler firesides, the supplier shall offer a choice of retractable lances, or of multi-nozzle elements suitably air cooled by gas baffles or air jackets for approval by the bureau or agency concerned. Suitable material and design to withstand the temperature will also be considered.

3.7 Bearings or hangers (for type I and type II soot blowers only).-

3.7.1 Purpose and position of bearings.- Bearings shall be provided to support the soot blower element and to preserve alignment. Ample clearance shall be provided in the bearings for the elements, to prevent seizing. The distance between bearings shall be not more than 2 feet 8 inches. When the axis of the element is at a right angle to that of the tubes, the bearing next to the boiler casing fitting shall be located with its center not more than six tubes from the end of the tube bank and the projection of the element at the closed end beyond the last bearing shall be not more than six inches. When the axis of the element is parallel to that of the tubes, the element shall be supported as approved by the bureau or agency concerned for the particular case. Material of bearings shall be the same as that of the element with which they are installed except that slight variations in chemical characteristics due to difference in fabrication methods will be permitted.

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3.7.2 Relation of bearings to tubes.- Unless otherwise approved by the bureau or agency concerned, the bearings shall be clamped to the tubes by U-bolts and nuts. When the slope of the tubes at the point of attachment of the bearings is greater than 30 degrees above the horizontal, the bearings shall be double clamped to decrease the possibility of slipping. Bearings shall contact at least three tubes to prevent turning. The surfaces of bearings in contact with tubes shall be machined accurately to fit the circumference and curvature of the tubes to obtain the maximum contact surface and reduce the temperature of the bearings by conduction heat to the tubes.

3.8 Boiler casing fitting.-

3.8.1 Boiler casing sleeve.- Suitable sleeves to allow free passage of the element or lance through the boiler casing is to be part of the design. If the supplier's design is such that this part may be fabricated by the boiler supplier or by the shipbuilder, the soot blower supplier is to be responsible for ensuring on each contract or order that the cognizant activity is advised of this requirement, including sketches and material list, to allow fabrication and installation.

3.8.2 Boiler casing seal or stuffing box.- The stuffing box shall be designed to permit passage of the rotary soot blower element through the boiler outer casing and to effectively seal the sleeve opening against leakage of combustion gases. This box shall also be designed to compensate for the relative movement between the connecting steam piping, the boiler tubes, and the boiler casing by a floating action which permits slight movement of the head in any direction in the plane normal to the axis of the element. Material for stuffing box may be either cast steel or ductile iron. Cast iron or malleable iron will not be permitted. Stuffing boxes shall be flanged. Provision shall be made for air connection to 1/2 inch o.d. tubing with entry to the stuffing box through at least 3/8 inch NPT connections.

3.9 Drawings.- Drawings shall be furnished in accordance with Specification MIL-D-963. Bids on soot blowers shall be accompanied by preliminary drawings showing the number and location, with respect to the tubes, of the nozzles in each element. An arrangement drawing showing the location of the soot blowers on the boiler as finally approved by the bureau or agency concerned shall be furnished. Master sets, consisting of not more than six drawings per set, shall be prepared for such class and type of soot blower for which a supplier desires approval. These masters shall include all possible variations in construction. Commercial type accessories such as air-motors and accompanying gear shall be shown on one dimensioned assembly and major detail drawing, together with a list of material identifying by part and manufacturer the major replacement parts. Radiographic and magnetic particle inspection markings shall also be delineated on one of the drawings. On any contract or order, the previously approved master drawings indicated above shall serve as the nucleous for drawing preparation. When only types I and II soot blowers are required on a contract or order, total number of applicable drawings shall not exceed 6; number of drawings to cover all types of soot blowers on a contract or order shall not exceed 12.

3.10 Additional marking.- Identification plates and other markings shall be furnished in accordance with Specification MIL-P-15024. These plates shall installed on and furnished as a part of the machinery or equipment for which they are intended.

3.10.1 Information plates.- Information plates containing instructions for operation and upkeep of the soot blowers shall be furnished. On each contract or order for application to new boilers, sufficent plates shall be furnished to service each different type of unit on each boiler. The plates shall be suitable for attachment to boiler casings near the applicable soot blower type installations. Plate material shall be suitable for 150 degree ambient temperature.

3.10.2 Identification of repair parts shall be in accordance with Specification MIL-P-15137.

3.10.3 Head marking.- All class 1 head castings shall include head marking in an easily discernible location. Marking shall be the designation "600" in figures at least 1/2 inch high. Class 2 head castings shall include the designation "1500" at least 1/2 inch high.

3.11 Repair parts and tools.-

3.11.1 Valve or head repair parts.- One set of valve or head repair parts shall be furnished for each ship on which the soot blowers are to be installed. Each set of repair parts shall consist of the following, the quantity to be 10 percent of the total number of soot blowers installed on the ship.

(a) Valve disc and stem sub-assembly.
(b) Pinion gear.
(c) Springs of each size used.
(d) Swivel tube (if applicable to design).
(e) Packing.
(f) Flange gaskets.
(g) Lance for type III and type IV soot blowers.
(h) Element for hottest boiler.
(i) Feed tube for type III and type IV soot blowers.

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3.11.1.1 Other repair parts. - When air-drive operation is required, sufficient motor parts (bearings, and similar repair parts) shall be provided to maintain the motor for one year. The parts furnished shall be interchangeable with industry and federal standards. In addition, two air-motors shall be furnished as shipboard repair parts on new construction for every 10 power driven soot blowers installed per ship. Packing for stuffing boxes and traveling carriage parts equal to 10 percent of installed quantities shall be furnished. Any other repair parts deemed necessary for one year's maintenance shall be offered for specific approval by the bureau or agency concerned.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First unit test. - One unit of each new design and construction shall be forwarded to the Naval Boiler and Turbine Laboratory, Philadelphia, Pa., for the tests specified in 4.6 and after approval of the basic design the tests will not be repeated, unless there are changes in the design and construction of the soot blowers. The tests shall be made to determine satisfactoriness of steam control, economy in the use of steam, facility of operation, simplicity in installation, effectiveness in removal of soot, ruggedness of construction, quality of workmanship, and the suitability of the unit for the use specified. The ability of the soot blower to meet the above requirements shall be demonstrated by an actual installation on a boiler at the Naval Boiler and Turbine Laboratory, Philadelphia, Pa.

4.3 Sampling for acceptance inspection. -

4.3.1 Lot. - All soot blowers of the same class and type presented for delivery at one time shall be considered a lot for purposes of acceptance inspection. All parts required to be eventually assembled together to form complete soot blowers shall be included in the lot. Also, all repair parts and tools shall be included in the lot.

4.3.2 Sampling for examination. - A random sample of soot blowers shall be selected from each lot, for the examination specified in 4.4, in accordance with Standard MIL-STD-105 at inspection level II. The acceptable quality level shall be 1.5 percent defective.

4.3.3 Sampling for tests. - A random sample of soot blowers shall be selected from each lot, for the tests specified in 4.5, in accordance with Standard MIL-STD-105 at inspection level L-8. The acceptable quality level shall be 1.5 percent defective: however, the sample sizes shall be as specified in Standard MIL-STD-105, and the acceptance number shall be zero until the AQL permits one or greater.

4.4 Examination. - Each of the sample soot blowers selected in accordance 4.3.2 shall be visually and dimensionally examined to determine conformance with the requirements of this specification as to material finish, workmanship, dimensions, components, fits, lubrication, marking, instruction plate, and repair parts and tools. Any soot blower in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective soot blowers in any sample exceeds the acceptance number for that sample, the lot shall be rejected. The interchangeability of the repair parts shall be verified by trial assembly. The suitability of each tool shall be established by a demonstration.

4.5 Lot acceptance tests. - Each of the sample soot blowers selected in accordance with 4.3.3 shall be tested as follows:

(a) Hydrostatic test (see 4.6.2).
(b) Radiographic and magnetic particle test (see 4.6.6).
(c) Scavenging valve test (see 4.6.3).

Failure to pass any of the tests shall be counted as a defect and the soot blower shall be rejected. If the number of defective soot blowers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6 Test procedures. -

4.6.1 Element material test. - The maximum gas temperature for which soot blower element materials are suitable shall be determined by tests at a laboratory satisfactory to the bureau or agency concerned. During these tests, each material under test shall be subject to progressively increasing gas temperatures for a sufficient period of time to determine the maximum temperature for which it is suitable when used as a soot blower element. The materials shall also be tested for strength at elevated temperatures by subjecting the sample element to air pressure while it is heated to the desired temperature. After initial approval of an element material for a specific temperature limitation, the

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element of material test need not be repeated if intented location is designed within the temperature limitation. A test report of the initial testing and a statement of the boiler supplier as to the boiler design temperature in the intended location will be acceptable.

4.6.2 Hydrostatic test. - All parts of the soot blowers to be furnished which are subject to line steam pressure shall be tested before shipment by hydrostatic pressure to one and one-half times the design pressure of the boiler on which they are to be installed.

4.6.3 Scavenging (air relief) valve hydrostatic test. - The valve shall be tested by hydrostatic pressure to 2000 p.s.i.

4.6.4 Shock test. - Shock tests shall be conducted in accordance with type A, grade I of Specification MIL-S-901. Test is not to be repeated after original approval of a specific design.

4.6.5 Vibration test. - Vibration tests shall be conducted in accordance with Standard MIL-STD-167. Test is not to be repeated after original approval of a specific design.

4.6.6 Radiographic and magnetic particle inspection. -

4.6.6.1 Radiographic inspection. - All pressure containing castings rated at 600 pounds and above shall be 100 percent radiographically inspected in accordance with Standard MIL-STD-271 and Publication NAVSHIPS 250-692-13. Sections less than 1-inch in thickness shall meet class 1 requirements of Publication NAVSHIPS 250-692-13. Sections with thickness 1-inch and over shall meet class 2 requirements of Publication NAVSHIPS 250-692-13.

4.6.6.2 Magnetic particle inspection. - All pressure containing forgings rated at 600 pounds and above shall be 100 percent magnetic particle inspected in accordance with Standard MIL-STD-271.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. -

5.1.1.1 Soot blower components. - Soot blower components, see 3.4, except 3.4(b) and (c), shall be preserved and packaged method I in accordance with the guide lines of Specification MIL-P-116. All external unpainted ferrous metal surfaces and all internal surfaces in contact with steam shall be coated with preservative compound conforming to grade 4 of Specification MIL-C-16173. Antifriction bearing shall be filled to operating capacity with normal operating lubricant. All openings shall be plugged, capped or sealed to prevent entrance of dirt and moisture.

5.1.1.2 Bearings (or hangers) and elements. - The bearings (or hangers) and elements shall be preserved and packaged method III in accordance with the guide lines of Specification MIL-P-116.

5.1.1.3 Repair parts. - Unless otherwise specified (see 6.1), repair parts (stock and on-board) shall be unit protected in accordance with the guide lines of Specification MIL-P-116. Parts shall be individually packaged unless used in sets or quantities greater than one.

5.1.2 Level C. - Soot blower components, see 3.4, except repair parts shall be preserved and packaged in accordance with the suppliers commercial practice.

5.2 Packing. - Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Soot blower components. -

5.2.1.1 Level A. - Soot blower components, packaged as specified, shall be packed in wood, cleated-plywood, wood-nailed and lock-corner boxes in accordance with Specifications PPP-B-601 (overseas type), and PPP-B-621 (class 2). Shipping containers shall have caseliners conforming to Specification MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. Shipping containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto. The gross weight of the shipping containers shall not exceed 200 pounds. Shipping containers exceeding 200 pounds shall be modified by the addition of skids in accordance with the applicable box specification.

5.2.1.2 Level B. - Soot blower components, packaged as specified, shall be packed in the boxes and the requirements specified in 5.2.1.1 except, wood-cleated plywood boxes shall be of the domestic type, and nailed and lock-corner boxes shall conform to class 1. Caseliners shall not be required. Closure shall be in accordance with the applicable box specification or appendix thereto.

5.2.2 Repair parts (on-board and stock). -

5.2.2.1 Level A. - Repair parts, packaged as specified, shall be packed in wood wirebound, fiberboard, wood-cleated, wood-cleated-plywood, wood-nailed and lock-corner, fiberboard and wood-cleated, veneer, paper overlaid boxes

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conforming to Specifications PPP-B-585 (class 3 use), PPP-B-591 (overseas type), PPP-B-601 (overseas type), PPP-B-621 (class 2), PPP-B-636 (class 2), and MIL-B-10377 (overseas type), as applicable at the option of the contractor. Shipping containers shall have caseliners conforming to Specification MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. Caseliners for class 2 fiberboard boxes conforming to Specification PPP-B-636 may be omitted provided all corners and edge seams and manufacturer's joints are sealed with minimum 1-1/2 inch wide tape conforming to Specification PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.2.2 Level B. - Repair parts shall be packed in any of the boxes specified in 5.2.2.1 and as follows:

(a) Caseliners shall not be required.
(b) Wirebound boxes shall be for class 1 or 2 use, wood or wood-cleated boxes shall be of the domestic type, and fiberboard boxes shall conform to class 2.
(c) Closure of boxes shall be in accordance with the applicable box specification or appendix thereto.
(d) The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.2.3 Repair parts boxes. - When specifically required in the contract or order, the on-board repair parts shall be packed in type M (metal) or type W (wood) boxes conforming to Specification MIL-B-233 as specified (see 6.1). Repair parts boxes will not require overpacking for domestic shipment or storage but shall be overpacked in containers specified in 5.2.2.1 for overseas shipment.

5.2.2.4 Index list of repair parts. - An index list of repair parts shall be inserted in each shipping container or repair parts box containing repair parts. The list shall contain a complete itemized listing of the box contents including stock numbers, nomenclatures and quantities. The list shall be placed in a waterproof bag or be so treated as to be resistant to water, oil and fading. The list shall be inserted in the index list support located on the interior side of the repair parts box cover, or suitably placed on the inside of the box for quick accessibility.

5.2.3 Anchoring, blocking, bracing and cushioning. - Anchoring, blocking, bracing and cushioning of the soot blower components and repair parts, when necessary, shall be provided to prevent damage during handling, shipment and storage.

5.2.4 Level C. - Soot blower components and repair parts, packaged as specified, shall be packed in a manner which will insure acceptance by common carrier and safe delivery at destination. Shipping containers or method of packing shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 Tools. - Tools shall be preserved, packaged, and packed for the level specified (see 6.1) in accordance with Specification MIL-P-15424.

5.4 Drawings. - Drawings shall be packaged and packed for the level specified (see 6.1) in accordance with Specification MIL-D-963.

5.5 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Types and classes required (see 1.2).
(c) Whether valve is required for type I soot blowers (see 3.5.2).
(d) Means of operation required for types II, III and IV soot blowers (see 3.5.3.2, 3.5.3.4 and 3.11.1.1).
(e) Levels of preservation, packaging and packing required (see 5.1 and 5.2).
(f) Type of repair parts boxes required (see 5.2.2.3).
(g) Levels of preservation, packaging and packing required for tools (see 5.3).
(h) Level of packaging and packing required for drawings (see 5.4).

NOTICE. - When Government drawings specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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